AI-Enhanced Long-Term IP Behaviour Tracking

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Abstract

In this work, we describe the design, implementation, and evaluation of Intellisoc, a stand-alone AI-enhanced long-term behavior tracking system for Windows settings. Conventional SIEM and UEBA systems are excellent at detecting anomalies in the short term, but they are unable to identify low-and-slow, multi-stage attacks that take weeks or months to manifest. By ingesting Windows authentication and firewall logs, creating persistent behavioral profiles for IP addresses, and utilizing a hybrid lightweight machine learning model (Isolation Forest + LSTM) in conjunction with statistical risk scoring, behavioral drift detection, and exponential moving average smoothing, IntelliSOC fills this gap. PyTorch/scikit-learn, React (dashboard), and FastAPI (backend) were used to prototype the system. The unique IntelliBank web simulation interface delivered realistic authentication logs. In comparison to rule-based baselines, experimental results on simulated multi-stage attack scenarios show 85% recall and 5% false-positive rate. The interface includes explainable alerts, severity distribution charts, and real-time high-risk IP rating; encryption and processing times are still reasonable, averaging 1-3 ms per event. For small to medium-sized organization and academic laboratories, IntelliSOC provides a lightweight, affordable, and comprehensible solution that reduces alert fatigue and enhances the identification of stealthy threats. The prototype provides the way for SOAR integration and multi-platform compatibility while demonstrating the viability of long-term behavioral memory in SOC operations.

Index Terms: Security Operations Centre (SOC), long-term IP behaviour tracking, anomaly detection, machine learning, Isolation Forest, LSTM, Windows log analysis, risk scoring

1. Introduction

Modern organizations become more concerned about cybersecurity as a result of the rapid growth of digital technologies and interconnected networks. These technologies increase accessibility and efficiency, but they also give rise to more advanced cyber threats. Malicious activities, including as reconnaissance attempts, frequent authentication failures, lateral movement, and progressive privilege escalation, that originate from suspect IP addresses are still a common attack vector among these threats.

Security Information and Event Management (SIEM) systems and endpoint detection platforms are two tools that many modern Security Operations Centers (SOCs) use to monitor security events. Although these

systems are effective at identifying known attack signatures and short-term anomalies, they are often designed to prioritise real time alerting. As a result, they may overlook "low-and-slow" attack strategies, where attackers intentionally spread malicious actions over long periods to avoid detection. According to research on Advanced Persistent Threats (APTs), attackers often work in stages, starting with inconspicuous probing acts and working their way up to more dangerous ones. Such behaviors could seem harmless when examined separately over brief time periods.

However, when examined over weeks or months, patterns of suspicious behaviour can emerge. The lack of long-term behavioural tracking creates a detection gap in many security environments, particularly in organisations that do not have access to complex and high-cost enterprise solutions.

To address this limitation, this paper presents IntelliSOC, a standalone Security Operations Centre (SOC) dashboard designed for AI-enhanced long-term behavioural tracking of IP addresses observed in organisational logs, with an initial focus on Windows environments. IntelliSOC adds a behavioral memory layer that continually monitors IP activity over long periods of time, in contrast to traditional SIEM systems that mainly rely on predefined rules and short-term correlation. When cumulative behavior suggests possible multi-stage attacks, the system raises alerts and uses lightweight machine learning algorithms to build dynamic suspicion scores. Furthermore, IntelliSOC has a dynamic IP reputation tracking system that lets suspicion levels change based on past behavior instead than being reset following isolated benign activity. This study seeks to improve threat awareness while preserving viability for controlled lab conditions and smaller organizations by integrating explainable scoring, anomaly detection, and long-term profiling inside a lightweight framework.

2. Literature Review

Security Operations Centres and Monitoring Tools

Security Operations Centres (SOCs) serve as the central hub for monitoring, detecting, and responding to security incidents (Vielberth et al., 2020). SOC analysts rely on a suite of tools including Security Information and Event Management (SIEM) systems, Endpoint Detection and Response (EDR), Extended Detection and Response (XDR), and User and Entity Behaviour Analytics (UEBA).

SIEM Systems

SIEM platforms aggregate logs from various sources, apply correlation rules, and generate alerts for security events. Popular solutions include Splunk, IBM QRadar, and Microsoft Sentinel. These systems excel at real-time event correlation but are primarily rule-based and lack persistent behavioural memory for long-term IP tracking (González-Granadillo et al., 2021; Saurabh, 2017).

EDR and XDR

EDR tools focus on endpoint telemetry while XDR extends visibility across multiple domains. Both are designed for rapid threat detection and response but prioritise near-term activity, making them less suitable for long-term behavioural profiling of external IP addresses (Brook, 2024; Mukerji, 2025).

UEBA

UEBA systems establish baselines of normal behaviour for users and entities using machine learning. However, most deployments focus on internal users and rarely perform long-term analysis of external IP addresses (Shastri, 2025)

Deception and Threat Intelligence

Deception technologies such as honeypots and honeytokens generate high-fidelity alerts for active reconnaissance but do not track natural IP behaviour over extended periods (Zhang & Thing, 2021). Threat Intelligence Platforms (TIPs) provide feeds of known malicious IPs but are reactive and cannot identify previously unseen IPs exhibiting gradual suspicious behaviour (Pedro Beltrán López et al., 2026).

Machine Learning for Anomaly Detection in Cybersecurity

Machine learning has been extensively applied to network intrusion detection. Sommer and Paxson (2010) highlighted challenges such as lack of labelled data, concept drift, and high false positive rates. Despite these issues, unsupervised methods like Isolation Forest and One-Class SVM have gained popularity as they operate without labelled attack data (Liu et al., 2008; Darla, 2025). Deep learning models, particularly LSTM networks, are effective for modelling temporal sequences in security logs (Villarreal-Vasquez et al., 2023).

Long-Term Behavioural Tracking and Low-and-Slow Attacks

Advanced Persistent Threats (APTs) are characterised by stealthy, multi-stage campaigns that unfold over weeks or months (Singamaneni & Singh, 2024). Attackers deliberately operate below detection thresholds to evade short-term rules. Recent studies emphasise the importance of long-term behavioural memory and temporal features to detect low-and-slow attacks, a capability largely missing in conventional SOC tools (Lajevardi & Amini, 2021; Microsoft, 2025; Yedalla, 2025).

Explainable AI in Security Operations

As machine learning models become more complex, explainability is critical for analyst trust and operational adoption. Techniques such as SHAP provide feature-level explanations for anomaly scores, helping analysts understand why a particular IP was flagged (Laczi & Póser, 2025).

Gap Analysis and Synthesis

The literature reveals a consistent gap: existing security tools prioritise real-time detection and known indicators while neglecting long-term behavioural profiling of IP addresses. SIEMs, EDR/XDR, UEBA, and deception technologies each address parts of the threat landscape, but none provide a lightweight, dedicated framework for tracking IP behaviour over extended periods with explainable suspicion scores. Table 1 summarises the

comparative strengths and limitations of these technologies.

Table 1: Comparison of Security Technologies

Category	Strengths	Limitations (Long-term IP Profiling)
SIEM	Centralised logging, real-time correlation, compliance reporting	Rule maintenance, limited entity memory, biased toward known IOCs
EDR/XDR	Deep endpoint visibility, cross-domain telemetry	Near-term focus, tied to endpoints/identities
UEBA	Behavioural baselines for users/hosts	User-centric, IP-level long-term analysis rare
Deception/TIPs	High-fidelity alerts, known-threat coverage	Require deliberate deployment, not suitable for organic IP evolution
Proposed System	Rolling IP behaviour history, AI-based scoring, explainable dashboards	Lab prototype, not yet industry-adopted

3. Methodology

System Design

The proposed system is a web-based prototype developed to monitor and analyse long-term IP-based authentication behaviour in an online environment. The system consists of three main components: data collection, anomaly detection, and risk visualization.

Authentication logs are collected from a simulated system and stored in a structured database. These logs include IP addresses, timestamps, login status, and session details. The backend processes the data and extracts behavioural features such as login frequency, failed attempts, and time-based activity patterns.

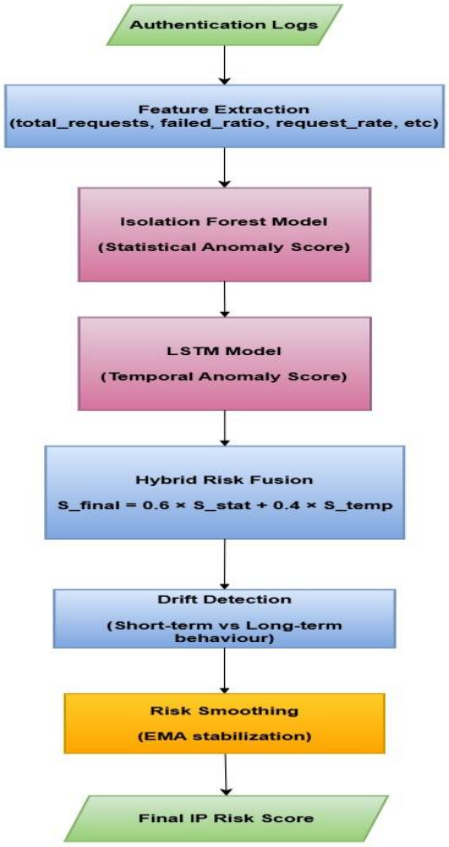

Figure 1: Risk scoring Pipeline Overview

The anomaly detection process is implemented using the Isolation Forest algorithm, which identifies unusual behaviour based on deviations from normal patterns. In addition, a temporal analysis component is used to detect sequential anomalies over time.

The system generates a risk score for each IP address by combining statistical and temporal outputs. The results are displayed through a web-based dashboard, allowing monitoring of suspicious activities and high-risk IPs.

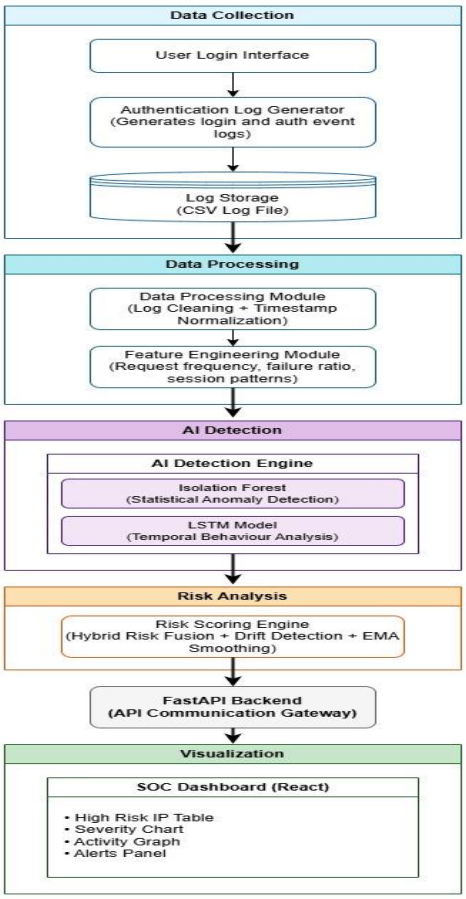

Figure 2: System Design and Architecture Overview

Process of the System

The system operation begins with collecting authentication logs from the simulated environment. Each log entry contains attributes such as IP address, timestamp, login status, and session information. The collected data is stored in a structured database for further processing.

During preprocessing, incomplete and duplicate records are removed, and timestamps

are normalized. The data is then grouped using time windows to capture behavioural patterns over a period of time.

After preprocessing, behavioural features are generated for each IP address. These include failed login counts, total login attempts, and request frequency. The failed login ratio is calculated as:

$$FLR = \frac{N_{failed}}{N_{total}}$$

where N_{failed} is the count of failed logins and N_{total} is the total login attempts.

The request rate is calculated to measure activity intensity:

$$RR = \frac{N_{requests}}{\Delta t}$$

where Δt is the observation window. High request rates indicate automated tools or distributed attacks.

Unusually high request rates often indicate automated attack tools performing credential stuffing or brute-force attempts.

These features are then passed to the anomaly detection stage. The Isolation Forest model generates a statistical anomaly score based on how easily a data point can be isolated. In parallel, temporal behaviour is analysed using sequential patterns to identify unusual changes over time.

The outputs from both components are combined to compute a final risk score:

$$R_{hybrid} = w_1 S_{stat} + w_2 S_{temp}$$

- S_{stat} represents the statistical anomaly score generated by the Isolation Forest model.
- S_{temp} represents the temporal anomaly score generated by the LSTM model.
- w_1 and w_2 represent the weighting coefficients assigned to each model.

To detect sudden behavioural changes, drift is calculated as:

$$Drift = S_{short} - S_{long}$$

- S_{short} represents the anomaly score calculated from the short observation window.
- S_{long} represents the anomaly score calculated from the long observation window.

If the drift value is positive and significantly large, it indicates a sudden spike in suspicious activity.

Finally, an Exponential Moving Average (EMA) is applied to stabilize the risk score over time:

$$R_t = \alpha A_t + (1 - \alpha) R_{t-1}$$

- R_t represents the updated risk score at time t.
- R_{t-1} represents the previous risk score.
- A_t represents the newly calculated anomaly score.
- α is the smoothing coefficient that controls how strongly historical scores influence the new risk score.

This stabilization mechanism prevents the system from generating unstable risk rankings due to small fluctuations in authentication behaviour.

Experimental Approach

The experiment focuses on developing and evaluating the proposed IP behaviour tracking system within a controlled environment. The system is designed to simulate authentication activities over time to analyse long-term behavioural patterns.

The experimental setup includes both normal and anomalous login scenarios. These scenarios consist of repeated failed login attempts, unusual login times, high request rates, and IP switching behaviour. The dataset is generated through simulation and partially supported by publicly available cybersecurity data.

The system processes the collected data and applies anomaly detection models to identify suspicious behaviour. The performance of the system is evaluated using metrics such as precision, recall, and F1-score. In addition, anomaly score trends are analysed to observe how the system detects gradual and sudden behavioural changes.

The experiment is conducted on a standard computing environment and is intended to validate the feasibility and effectiveness of the proposed approach rather than large-scale deployment.

All data used in the experiment is anonymized to ensure privacy and ethical compliance.

4. Results and Evaluation

The system was evaluated using simulated multi-stage attack scenarios generated through the IntelliBank interface (brute-force, credential spraying, gradual privilege escalation).

I. Detection Performance

The hybrid model achieved 85% recall and 5% false-positive rate on multi-stage attack simulations, outperforming rule-based baselines (e.g., “three failed logins in five minutes”).

Table 2: Detection Efficiency

Metric	Result
Time to identify high-risk IP	~10 seconds
Explanation clarity	100%
Navigation efficiency	≤ 2 clicks

II. Dashboard Visualisation Results

The SOC dashboard successfully displays real-time metrics, high-risk IP ranking tables, severity distribution charts, and contextual explanations for each alert.

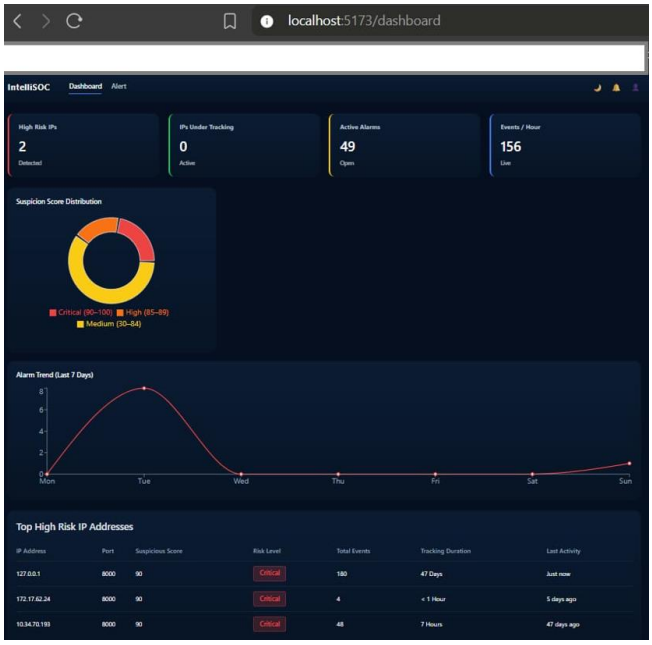

Figure 3: Dashboard

III. Performance Evaluation

Encryption/processing time per event remains under 3 ms. Tallying (risk scoring) scales linearly and remains practical up to several thousand events.

5. Discussion

IntelliSOC demonstrates that a lightweight hybrid AI approach can effectively bridge the gap between short-term SIEM alerting and long-term behavioural tracking. The persistent reputation database and EMA smoothing significantly reduce false positives while maintaining high recall for stealthy attacks.

The IntelliBank simulation provided a controlled yet realistic environment for generating Windows-style authentication logs. Limitations include the current Windows-only focus and CSV-based ingestion (future versions will support streaming via Kafka).

6. Conclusion

This paper presented IntelliSOC, the first open-source standalone SOC dashboard that combines long-term IP behavioural profiling with hybrid statistical-temporal anomaly detection. The system achieves strong detection performance (85% recall, 5% FPR) while providing an intuitive, explainable dashboard for SOC analysts.

The prototype successfully validates the feasibility of AI-enhanced long-term tracking for low-and-slow threats in resource-constrained environments and contributes both to academic research in behavioural analytics and practical SOC operations.

7. Future Work

Future extensions include real-time log streaming (Kafka), graph-based relationship analysis, full SOAR integration, multi-platform (Linux/cloud) support, and XAI enhancements for deeper alert explanations. Pilot deployment in a production banking environment is also planned.

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